



Apache Pig Built in Functions

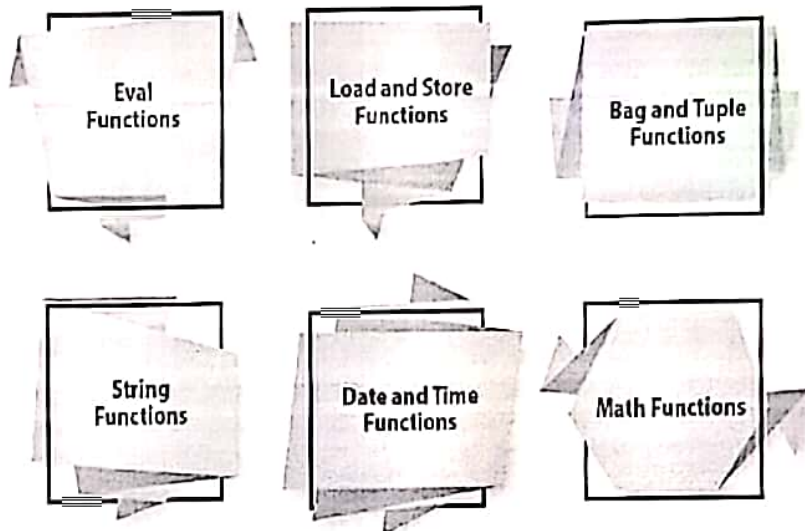
There is a huge set of Apache Pig Built in Functions available. Such as the eval, load/store, math, string, date and time, bag and tuple functions. Basically, there are two main properties which differentiate built in functions from user-defined functions (UDFs) such as:

- We do not need to register built in functions since Pig knows where they are.
- Also, we do not need to qualify built in functions, while using them, because again Pig knows where to find them.



Apache Pig Built in Functions

Pig Built in Functions





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Eval Function

S.N.	Function & Description
1	<u>AVG()</u> To compute the average of the numerical values within a bag.
2	<u>BagToString()</u> To concatenate the elements of a bag into a string. While concatenating, we can place a delimiter between these values (optional).
3	<u>CONCAT()</u> To concatenate two or more expressions of same type.
4	<u>COUNT()</u> To get the number of elements in a bag, while counting the number of tuples in a bag.
5	<u>COUNT_STAR()</u> It is similar to the COUNT() function. It is used to get the number of elements in a bag.
6	<u>DIFF()</u> To compare two bags (fields) in a tuple.
7	<u>IsEmpty()</u> To check if a bag or map is empty.



Eval Function

8	<u>MAX()</u> To calculate the highest value for a column (numeric values or chararrays) in a single-column bag.
9	<u>MIN()</u> To get the minimum (lowest) value (numeric or chararray) for a certain column in a single-column bag.
10	<u>PluckTuple()</u> Using the Pig Latin <code>PluckTuple()</code> function, we can define a string Prefix and filter the columns in a relation that begin with the given prefix.
11	<u>SIZE()</u> To compute the number of elements based on any Pig data type.
12	<u>SUBTRACT()</u> To subtract two bags. It takes two bags as inputs and returns a bag which contains the tuples of the first bag that are not in the second bag.
13	<u>SUM()</u> To get the total of the numeric values of a column in a single-column bag.
14	<u>TOKENIZE()</u> To split a string (which contains a group of words) in a single tuple and return a bag which contains the output of the split operation.



Load & Store Functions

S.N.	Function & Description
1	<u>PigStorage()</u> To load and store structured files.
2	<u>TextLoader()</u> To load unstructured data into Pig.
3	<u>BinStorage()</u> To load and store data into Pig using machine readable format.
4	<u>Handling Compression</u> In Pig Latin, we can load and store compressed data.



Bag & Tuple Functions

S.N.	Function & Description
1	<u>TOBAG()</u> To convert two or more expressions into a bag.
2	<u>TOP()</u> To get the top N tuples of a relation.
3	<u>TOTUPLE()</u> To convert one or more expressions into a tuple.
4	<u>TOMAP()</u> To convert the key-value pairs into a Map.



String Functions

S.N.	Functions & Description
1	<u>ENDSWITH(string, testAgainst)</u> To verify whether a given string ends with a particular substring.
2	<u>STARTSWITH(string, substring)</u> Accepts two string parameters and verifies whether the first string starts with the second.
3	<u>SUBSTRING(string, startIndex, stopIndex)</u> Returns a substring from a given string.
4	<u>EqualsIgnoreCase(string1, string2)</u> To compare two strings ignoring the case.
5	<u>INDEXOF(string, 'character', startIndex)</u> Returns the first occurrence of a character in a string, searching forward from a start index.
6	<u>LAST INDEX OF(expression)</u> Returns the index of the last occurrence of a character in a string, searching backward from a start index.
7	<u>LCFIRST(expression)</u> Converts the first character in a string to lower case.



String Functions

8	<u>UCFIRST(expression)</u> Returns a string with the first character converted to upper case.
9	<u>UPPER(expression)</u> UPPER (expression) Returns a string converted to upper case.
10	<u>LOWER(expression)</u> Converts all characters in a string to lower case.
11	<u>REPLACE(string, 'oldChar', 'newChar');</u> To replace existing characters in a string with new characters.
12	<u>STRSPLIT(string, regex, limit)</u> To split a string around matches of a given regular expression.
13	<u>STRSPLITTOBAG(string, regex, limit)</u> Similar to the STRSPLIT() function, it splits the string by given delimiter and returns the result in a bag.
14	<u>TRIM(expression)</u> Returns a copy of a string with leading and trailing whitespaces removed.
15	<u>LTRIM(expression)</u> Returns a copy of a string with leading whitespaces removed.
16	<u>RTRIM(expression)</u> Returns a copy of a string with trailing whitespaces removed.



Date-time Functions

S.N.	Functions & Description
1	<u>ToDate(milliseconds)</u> This function returns a date-time object according to the given parameters. The other alternative for this function are ToDate(iosstring), ToDate(userstring, format), ToDate(xserstring, format, timezone)
2	<u>CurrentTime()</u> returns the date-time object of the current time.
3	<u>GetDay(datetime)</u> Returns the day of a month from the date-time object.
4	<u>GetHour(datetime)</u> Returns the hour of a day from the date-time object.
5	<u>GetMilliSecond(datetime)</u> Returns the millisecond of a second from the date-time object.
6	<u>GetMinute(datetime)</u> Returns the minute of an hour from the date-time object.
7	<u>GetMonth(datetime)</u> Returns the month of a year from the date-time object.



Date-time Functions

8	<u>GetSecond(datetime)</u> Returns the second of a minute from the date-time object.
9	<u>GetWeek(datetime)</u> Returns the week of a year from the date-time object.
10	<u>GetWeekYear(datetime)</u> Returns the week year from the date-time object.
11	<u>GetYear(datetime)</u> Returns the year from the date-time object.
12	<u>AddDuration(datetime, duration)</u> Returns the result of a date-time object along with the duration object.
13	<u>SubtractDuration(datetime, duration)</u> Subtracts the Duration object from the Date-Time object and returns the result.
14	<u>DaysBetween(datetime1, datetime2)</u> Returns the number of days between the two date-time objects.



Date-time Functions

15	<u>HoursBetween(datetime1, datetime2)</u> Returns the number of hours between two date-time objects.
16	<u>MillisecondsBetween(datetime1, datetime2)</u> Returns the number of milliseconds between two date-time objects.
17	<u>MinutesBetween(datetime1, datetime2)</u> Returns the number of minutes between two date-time objects.
18	<u>MonthsBetween(datetime1, datetime2)</u> Returns the number of months between two date-time objects.
19	<u>SecondsBetween(datetime1, datetime2)</u> Returns the number of seconds between two date-time objects.
20	<u>WeeksBetween(datetime1, datetime2)</u> Returns the number of weeks between two date-time objects.
21	<u>YearsBetween(datetime1, datetime2)</u> Returns the number of years between two date-time objects.



Math Functions

S.N.	Functions & Description
1	<u>ABS(expression)</u> To get the absolute value of an expression.
2	<u>ACOS(expression)</u> To get the arc cosine of an expression.
3	<u>ASIN(expression)</u> To get the arc sine of an expression.
4	<u>ATAN(expression)</u> This function is used to get the arc tangent of an expression.
5	<u>CBRT(expression)</u> This function is used to get the cube root of an expression.
6	<u>CEIL(expression)</u> This function is used to get the value of an expression rounded up to the nearest integer.



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7	<u>COS(expression)</u> This function is used to get the trigonometric cosine of an expression.
8	<u>COSH(expression)</u> This function is used to get the hyperbolic cosine of an expression.
9	<u>EXP(expression)</u> This function is used to get the Euler's number e raised to the power of x.
10	<u>FLOOR(expression)</u> To get the value of an expression rounded down to the nearest integer.
11	<u>LOG(expression)</u> To get the natural logarithm (base e) of an expression.
12	<u>LOG10(expression)</u> To get the base 10 logarithm of an expression.
13	<u>RANDOM()</u> To get a pseudo random number (type double) greater than or equal to 0.0 and less than 1.0.



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Math Functions

14	<u>ROUND(expression)</u> To get the value of an expression rounded to an integer (if the result type is float) or rounded to a long (if the result type is double).
15	<u>SIN(expression)</u> To get the sine of an expression.
16	<u>SINH(expression)</u> To get the hyperbolic sine of an expression.
17	<u>SQRT(expression)</u> To get the positive square root of an expression.
18	<u>TAN(expression)</u> To get the trigonometric tangent of an angle.
19	<u>TANH(expression)</u> To get the hyperbolic tangent of an expression.